

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
COLLEGE OF ENGINEERING AND TECHNOLOGY, KATTANKULATHUR CAMPUS
DEPARTMENT OF ELECTRONICS AND INSTRUMENTATION ENGINEERING



Course Code	\$21EIE552T	Course Website:	
Course Title	CYBERSECURITY IN INDUSTRIAL AUTOMATION	Instructor:	Dr. G.Y.RAJAA VIKHRAM
Semester	EVEN 2025	Email:	rajaaviy@srmist.edu.in
Pre-requisite / Co-Requisite	NIL	Slot & Class Hours:	Slot A Day Order 3,4 and 5
Lecture Hall #	PROCESS CONTROL SYSTEM LAB	Laboratory Location	NA

Organisation of the course

Classes will be predominantly lectures with technology advancements and discussion on R&D works. Guest lectures from R&D will be invited.

Course Objectives and Learning Outcome(s)

- 1 Summarize the basic concepts of industrial automation and control systems
- 2 Illustrate any application with add-on security features in industrial control systems
- 3 Develop the security methodologies for industrial systems

Learning Materials

1. Steve Mustard, "Industrial Cybersecurity: Case Studies and Best Practices", ISA, 2022.
2. Charles J. Brooks, Philip A. Craig Jr. "Practical Industrial Cybersecurity", Wiley, 2022.
3. K S Manoj, Cyber Security : in industrial automation, Notion Press, 2020.
4. Ronald L. Krutz, "Industrial Automation and Control System Security Principles: Protecting the Critical Infrastructure", 2nd ed., ISA, 2017.
5. Perry S. Marshall. "Industrial Ethernet", ISA, 2017.
6. Edward J.M. Colbert, "Cyber Security of SCADA and other Industrial Control Systems", Springer, 2016

Course Topics and

Week	Major Topic	Module / Unit	Method of delivery (Multi-Select)	Take home Assignment(s)	Learning materials to be referred	CO	Related PO
1	IACS – Introduction - types of automation - working principles - applications - security issues in automation	1	Lecture		1	1	1
2	SCADA systems - Distributed control systems - Safety instrumented systems IACS	1	Lecture		1,2	1	1
3	Protocol - OSI model - TCP/IP model - OPC for process control - TCP, DNP3 protocol - Utility communication architecture	1	flipped class		1,2	1	1
4	Profibus - Controller area network - Ethernet/IP - Open safety protocol	1	Lecture		1	1	1



5	Issues in IACS security - Information security - Approaches of information security - Applications	1	virtual lab experiments	Assignment - 1	1	1	1
6	Information system security fundamentals - Terminologies - Threat matrix - Type and classes of attack	2	virtual lab experiments		2	2	1
7	Additional system security - Policies, standards - Guidelines and procedures - Malicious codes and attacks - Firewalls - Cryptography - Digital signatures	2	flipped class		2	2	1
8	Virtual private network - IPsec - Transport mode - Tunnel mode - Secure sockets layer - Physical and economic damage	2	Lecture		2,3	2	1
9	Differences in culture, philosophy, and requirements - Comparison between IT and IACS issues - Considerations in adapting IT security methods to IACS	2	virtual lab experiments		2	2	1,3
10	Threats and motivations for attackers - Threat sources - Technological trends - Smart grid and its trends - Smart grid protocols - Mapping of emerging technology - Example automation system	2	flipped class	Assignment - 2	2,3	2	1,3
11	Risk management - Risk relationships - Cybersecurity management systems - Monitoring and improving the IACS - Integrated enterprise risk management	3	Lecture		3,4	3	1,3
12	Guide for applying risk management framework - Insider threat - Threat example - Stuxnet - Defensive approaches - Electromagnetic pulse - HEMP - Protection measures -	3	Lecture		3,5	3	1,3
13	Standards - IEC & IEEE Cyber security lifecycle - Conceptual design process - Firewall design - Intrusion detection design	3	flipped class	Assignment - 3	3,6	3	1,3

	Security standards, Guidelines - NIST – Guide to ICS security, Management controls - ANSI/ISA security technologies, - Encryption technologies, Physical security controls - NERC – Critical infrastructure protection - Security management controls				3,6		
14		3	Lecture			3	1,3
15	Case study	3	Lecture	Conference / Journal publication	3	3	1,3

Grading Criteria

Type of assessments	Periodicity	Weightage	Due Date (if applicable)	Remarks	
FT1-20 - Test, CT - 1 - w	after completion of module 1. (after week 4)	20%	17-21/2/2026	Written exam	
written test = 20marks, Assignment - 2 - 5 marks, Seminar - 5 marks	after completion of module 2 and 3. (after week 12)	30%	21-25/4/2026	Written exam	
LLT - 1-10 marks, Conference paper acceptance = 10 marks.	towards the end of the semester.	10%	end sem	Conference / journal paper acceptance is must	
Continuous Assessment	Total	60%			
End Semester Examination		40%			
	Total	100%			

Make-up Policy

Assignments / Term Project

Proper academic performance depends on students doing their work not only well, but on time. Accordingly, assignments for this course must be received on the due date specified for the assignment. No Make-up will be available for assignments or term project. Late submissions will be evaluated at 25% less weight for that component for a delay of up to 24 hours after which no submissions will be accepted.

Test

Prior Permission of the course coordinator is usually required to get a make-up test.

A make-up test shall be granted only in genuine cases where, in the course faculty's judgment, the student would be physically unable to appear for the test. Course faculty's decision in this matter would be final.

Academic Integrity

Plagiarism or cheating will result in a grade of zero for the assignment or exam. Severe cases may result in course failure.

Additional Information

Provide a brief description (e.g. field trips, special lab session, special tutorials, industry certifications etc), dates, times, required materials or preparation, any fees or costs, etc.

Additional academic support

If any student requires additional free academic support (additional coaching), he/she may reach out to the Faculty Advisor

Notices

All notices concerning this course will be displayed online only. If there is a need srmist email would be used.

Contact Information

Feel free to reach out to Dr. Rajaa Vikhram(rajaaviy@srmist.edu.in) via email for any questions regarding the course material. For urgent matters, you can visit the faculty room during offi

Rajaa Vikhram
13/11/26

[Signature]